Membership versus variety, and the partial-solution strata: what changes if the PDE is added to the system before decomposition

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A companion note to *A New Method of Solving PDEs*. The core algorithm *reduces* the PDE P modulo the ansatz A and projects the remainder onto the constants, returning the locus $\mathcal{V}$ of constants for which the ansatz family solves the PDE. A natural-looking alternative is to adjoin P to the ansatz and run a differential decomposition (Rosenfeld–Gröbner or, better, differential Thomas) on the *combined* system $A \cup \{P\}$. This note records why that answers a *different* question, names the discrepancy — the *partial-solution strata* — and exhibits it on a two-line example.

1 Two different loci

Let $A(c) = \{a_1, \ldots, a_k\}$ be the ansatz, a differentially triangular set whose coefficients involve constant parameters $c \in \mathbb{C}^n$, and let P be the PDE. The core method computes the **membership locus**

$$\mathcal{V} \;=\; \{\, c \in \mathbb{C}^n : P \in [A(c)]\,\}, \tag{1}$$

a *universal* condition: *every* function in the ansatz family at c solves the PDE. Operationally, Ritt's full reduction of P modulo A yields a remainder $r = \sum_\alpha m_\alpha\, p_\alpha(c)$ with the m_α distinct power products in the non-constant indeterminates; r vanishes *identically* iff every coefficient $p_\alpha(c)$ vanishes, so $\mathcal{V} = V(p_1, \ldots, p_N)$.

Decomposing the combined system computes instead the **joint solution variety**

$$\mathrm{Sol}(A) \cap \mathrm{Sol}(P), \tag{2}$$

an *existential* condition: there *exists* a function satisfying both A and P. Projecting (2) onto the constants gives

$$\pi_c\big(\,\mathrm{Sol}(A) \cap \mathrm{Sol}(P)\big) \;=\; \mathcal{V} \;\cup\; \{\text{partial-solution strata}\} \;\supseteq\; \mathcal{V}. \tag{3}$$

Definition 1 (partial-solution stratum) *A* partial-solution stratum *is a set of constant values c at which a* proper, nontrivial *sub-family of the ansatz solutions satisfies P, while the family as a whole does not. Equivalently: $c \notin \mathcal{V}$, yet* $\mathrm{Sol}(A(c)) \cap \mathrm{Sol}(P)$ *contains more than the zero solution.*

A prerequisite is immediate from Definition 1: the ansatz solution space at c must be at least two-dimensional. A one-dimensional family either solves P in its entirety or only through its zero member; there is no proper nontrivial sub-family to peel off.

2 A two-dimensional ansatz where the strata appear

Example 1 *Take a two-dimensional ansatz and a first-order PDE,*

$$A(c): \quad \Psi'' - c\,\Psi = 0, \qquad\qquad P: \quad \Psi' - \Psi = 0. \tag{4}$$

For $c = \omega^2$ the ansatz solution space is two-dimensional, $\Psi = K_1 e^{\omega x} + K_2 e^{-\omega x}$.

Membership (core method): reduce P modulo A. The leader of P is Ψ', which is ranked *below* the leader Ψ'' of A, so P is already fully reduced with respect to A; the remainder is P itself,

$$r \;=\; \Psi' - \Psi \;=\; (+1)\cdot\Psi' + (-1)\cdot\Psi.$$

The projection coefficients are the constants $+1$ and -1: never zero, with no c-dependence. Hence

$$\mathcal{V} \;=\; V(1,-1) \;=\; \varnothing.$$

This is correct: no value of c makes the entire two-parameter family $\{K_1 e^{\omega x} + K_2 e^{-\omega x}\}$ obey the first-order condition $\Psi' = \Psi$.

Combined decomposition: adjoin P, rank the constant lowest. The decomposition selects the lower-leader equation $\Psi' - \Psi$ into the triangular set first, then reduces $\Psi'' - c\,\Psi$ against it. Using $\Psi' = \Psi \Rightarrow \Psi'' = \Psi$,

$$\Psi'' - c\,\Psi \;\longrightarrow\; \Psi - c\,\Psi \;=\; (1-c)\,\Psi \;=\; 0.$$

This equation has leader Ψ and initial $(1-c)$, so the algorithm splits on the vanishing of $(1-c)$:

cell	reduced system	solutions
$c \neq 1$	$(1-c)\Psi = 0, \quad \Psi' = \Psi$	$\Psi \equiv 0$ (trivial only)
$c = 1$	$0 = 0, \quad \Psi' = \Psi$	$\Psi = Ke^x$ (one-parameter)

The cell $c = 1$ is the partial-solution stratum. At $c = 1$ the ansatz has the two modes e^x and e^{-x}; the PDE $\Psi' = \Psi$ retains the e^x mode and discards e^{-x}. Exactly half the family solves the PDE — "some, but not all" — so $c = 1 \notin \mathcal{V}$ even though it is a genuine, nonempty solution stratum of the joint system.

3 The correct selection criterion

Equation (3) warns that projecting the *whole* combined variety over-counts $\mathcal{V}$. One might hope to recover $\mathcal{V}$ by keeping the cell in which the adjoined PDE-equation collapsed to $0 = 0$; Example 1 shows this is wrong, as that cell is $c = 1$ whereas $\mathcal{V} = \varnothing$. The collapse cell is the *existential* locus, not the membership locus.

The correct criterion is **dimension preservation**: $c \in \mathcal{V}$ iff the combined solution space at c still equals the *full* ansatz solution space, i.e. the PDE removed nothing. In Example 1 no cell retains the full two-dimensional space (the $c = 1$ cell keeps only a one-dimensional slice), so $\mathcal{V} = \varnothing$, consistent with the direct reduction.

4 Contrast: a one-dimensional ansatz has no partial strata

Shrink the ansatz to one dimension and test a second-order PDE:

$$A(c): \quad \Psi' - c\,\Psi = 0 \quad (\Psi = Ke^{cx}), \qquad\qquad P: \quad \Psi'' - \Psi = 0.$$

Now P's leader Ψ'' dominates A's leader Ψ'. Reducing P modulo A gives $\Psi'' \to c^2\Psi$, hence $r = (c^2 - 1)\,\Psi$ and

$$\mathcal{V} \;=\; V(c^2 - 1) \;=\; \{\, +1,\, -1\,\}.$$

The combined decomposition produces the cells

$$\{\, c^2 \neq 1 : \Psi \equiv 0\,\}, \qquad \{\, c = 1 : \Psi = Ke^{x}\,\}, \qquad \{\, c = -1 : \Psi = Ke^{-x}\,\},$$

and the nontrivial cells $c = \pm 1$ now retain the *entire* (one-dimensional) family. Here the collapse cells coincide with $\mathcal{V}$ and there are no partial strata: a one-dimensional family admits no proper nontrivial sub-family.

5 Reading

The membership formulation (1) asks: *for which ODE-coefficients c is the PDE an automatic consequence of the ansatz?* — so that the ansatz family is, as a whole, a PDE-solution family. The partial strata answer a different question: *for which c does the ansatz family happen to contain a PDE solution it was not designed to?* — coincidental solutions living on a proper sub-family at special parameter values. They are real solutions, but invisible to the membership formulation by construction.

For the hydrogen computation this means: adjoining the Schrödinger equation to the ansatz and decomposing surfaces parameter values where a degenerate, lower-dimensional slice of the ansatz solves the PDE while the generic member does not. If the goal is the membership locus $\mathcal{V}$, either reduce P modulo A directly, or — having decomposed the combined system — keep only the dimension-preserving cells. Projecting the full joint variety answers the existential question and returns $\mathcal{V}$ together with the partial strata.